

CHAPTER THIRTY-ONE

Worlding Systems Theory

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We live in strange times, a little as if we were suspended between two histories, both of which speak of a world become “global.”

—Isabelle Stengers, *In Catastrophic Times: Resisting the Coming Barbarism*

The world of modern society is a background indeterminacy (“unmarked space”) that allows objects to appear and subjects to act.

—Niklas Luhmann, *Theory of Society, Volume 1*

This circularity, this connection between action and experience, this inseparability between a particular way of being in the world and how the world appears to us, tells us that *every act of knowing brings forth a world*.

—Humberto Maturana and Francisco Varela, *The Tree of Knowledge*

So that is the problem. Whether to assume the world is one and that we are all inside it; or, instead, to wrestle with the implications that worlds in the plural are enacted in different and power-saturated practices.

—John Law, “What’s Wrong with a One-World World?”

Conceptual analysis can carry us only partway in the journey toward more relational living. To the theoretical work we need to add some form of practice that takes us into other habits and modes of living and interexisting, of being in a world that is made up of things that are real yet not inherently independent.

—Arturo Escobar, *Designs for the Pluriverse*

A world philosophy based on the recognition of the autopoietic and nonmechanical nature of life must upset the believers in the fundamental myths of our technological civilization.

—Lynn Margulis, “Big Trouble in Biology”

This chapter addresses systems theory in its second-order line of development—*neocybernetic systems theory* (NST). NST takes off from the second-order or recursive turn in the cybernetics of Heinz von Foerster and the calculus of distinctions in George Spencer-Brown’s *Laws of Form*. *Laws of Form* reinforced von Foerster’s discourse of cognitive recursion with a seminal exposition of *reentry*, the “reentry of the form into the form.” By such self-referential processes, systems may elaborate and temporalize their internal complexity. Von Foerster

was also instrumental in disseminating the concept of autopoiesis developed in the biological systems theory of Humberto Maturana and Francisco Varela. Here, recursion and reentry are instantiated by the circular operation of a self-producing system—foundationally, a living cell in its continuous metabolic self-maintenance. An autopoietic system cuts itself out of (but not off from) its environment by constituting its own operational boundaries, within which it issues forth as the product of its own production. Niklas Luhmann then radicalized the epistemological constructivism that von Foerster drew from his analysis of recursive closure, operationalized Spencer-Brown's formal calculus of distinctions, and lifted the conceptual architecture of autopoiesis out of its original biological reference, incisively reformatting them within his theory of social systems as composed of communications (rather than persons). For Luhmann, both psychic and social systems are metabiotic autopoietic systems structurally coupled by their mutual reliance (for the discrete operations of thinking and of communicating) on the medium of meaning carried along by events of distinction marked by semiotic structures. I have treated NST at some length in previous publications and refer my reader to them for basic historical and conceptual details.¹

Within NST, social systems theory also develops a concept of *world society*. The world that modernity brings forth is the final horizon of meaningful operations for this “encompassing social system.”² What world society encompasses in the first instance are the function systems that have achieved semi-autonomous operation as discrete social subsystems—the legal system, the economic system, the political system, and so on—that take world society for their common environment. Comparisons with alternative conceptions of the world in John Law's amalgamation of science and technology studies (STS) and actor-network theory (ANT), as well as in Arturo Escobar's systems-theoretical approach to design theory, will provide critical perspectives on the world corresponding to world society in social systems theory. We will then return to consider Luhmann's own presentation of how functional differentiation in world society generates inequality and social exclusion as a matter of course. The theory states that social systems evolve on their own terms and cannot be steered by their environments toward intentional outcomes. Nevertheless, even in Luhmann's text there are margins of indeterminacy at the edges of systems. Given this unpredictable interplay of operational closure and environmental openness, it may be said that even if social systems cannot be steered, they can be nudged. And once we take autopoiesis back to its worldly instantiation in living systems, world society may be placed in planetary context as open to the psychic processing of ecological noise, or, in the phrase of Isabelle Stengers, the intrusion of Gaia. This newly insistent recognition of environmental agency, codified in the work of Lynn Margulis as autopoietic Gaia, has the potential to shake enough sense into the function systems to prolong world society's run past what otherwise appears to be a near-term expiration date.

WORLD SOCIETY

In his introduction to *Social Systems* (German original, 1984), Luhmann notes that “The following investigations hold strictly to the level of a general theory of social systems. They do not, for example, offer a theory of society—society understood as a comprehensive social system.”³ The theory of social systems is not yet the theory of society. *Social Systems* is the substantial general prolegomenon for which Luhmann's final work, *Die Gesellschaft der Gesellschaft* (German original, 1997), is the fulfillment. The literally recursive German title, “the society of society,” is tamed or rendered linear in English translation as *Theory of Society*.

So although there are countless instances of social systems—the minimal formation being an “interaction system” constituted by the occurrence of communication between two distinct addressees (say, a conversation between two persons)—under globalized modern conditions, there is ultimately only one society, that is, *world society*, in relation to which the ongoing differentiations of all other social systems are subsystemic developments.

Modern society is world society. “It is the encompassing social system which includes all communications, reproduces all communications and constitutes meaningful horizons for further communications. Society makes communication between other social systems possible. Society itself, however, cannot communicate. Since it includes all communication, it excludes external communication. It has no external referent for communicative acts, and looking for partners would simply enlarge the societal system.”⁴ For instance, if human beings could suddenly communicate with dolphins or extraterrestrial beings by exchanging meaningful utterances with them, these nonhuman communication partners would relinquish their social alterity. While the environment of world society would assume a more complex texture, their utterances would simply be added to and so enlarge world society, prompting its further evolution. If developments in machine intelligence yield what may be characterized as events of communication between human beings and computational agencies, the producers of these “artificial” utterances must also take their place alongside human beings in the environment constituted by world society as the encompassing social system.⁵

With all due respect to Bruno Latour’s influential declaration that *we have never been modern*, in Luhmann’s analysis of the evolution of society, it does not matter what “we” (persons) may be or may (never) have been. Rather, *society was not always modern, but now it is*. In modern society, the stratified form of premodern societies—differentiated by class, rank, birthright, and other hierarchical structures—has definitively given way to the form of functional differentiation. Vestiges of concepts germane to social stratification remain in the antique gallery of ideas, such as the “oppression” by “upper” of “lower” social classes. However, for world society, the social production of inequalities now arises from the altered dynamics generated by an uncoordinated array of semiautonomous function systems, and the possibility of redressing grievances through structural reforms depends on registering this observation and adapting one’s strategies accordingly.

With world society, the vertical orientation of social stratification withdraws before the horizontality of functional differentiation. An archetype of this development is the separation of church and state at the foundation of modern secularity. For modern society, “function, not rank, is the dominant principle of system building.”⁶ The function systems of world society include the economic subsystem, the political subsystem, the legal subsystem, the educational subsystem, the scientific subsystem, the media subsystem, and the art subsystem. All function systems operate and evolve on their own terms, more specifically, in terms of their own dedicated codes—for instance, the economic code of payment/non-payment, the legal code of lawful/unlawful, the media code of information/non-information, and so on. Each function system draws boundaries around the determination of its own communications. Only the legal system can decide what is lawful and what is not, while in the economic system, lawfully or not, money talks, and only money (or the promise of its arrival) talks.

As Luhmann writes in “World Society,”

The inclusion of all communicative behavior into one societal system is the unavoidable consequence of functional differentiation. Using this form of differentiation, society becomes a global system ... It includes all human (i.e. meaningful) communication. Modern

society is, therefore, a world society in a double sense. It provides one world for one system: and it integrates all world horizons as horizons of one communicative system. The phenomenological and the structural meanings converge. A plurality of possible worlds becomes inconceivable.⁷

Reviewing these forty-year-old remarks, it is plain that a “plurality of possible worlds becomes inconceivable” strictly in relation to the stringent autopoietic logic to which Luhmann adheres throughout his theoretical project. In the section of *Theory of Society* on “World Society,” Luhmann returns to this theme and confirms his observation with regard to it:

The consequence of defining society as the comprehensive social system is that there can be only one system of society for all connective communication. In purely factual terms, a number of societal systems may well exist, just as one used to speak of a plurality of worlds; but only in the absence of communicative links between these societies or, from the point of view of the individual society, where communication with others is impossible or without consequence.⁸

The math for Luhmann’s theory of world society is straightforward: social systems are plural, but society is singular. The number of all possible social systems is without upper limit; the number of function systems is similarly an open figure; but the number of societies is one, and world society is it. In concrete terms, on this planet there may still exist—for instance, in the depths of the Amazon forest—social enclaves standing apart from world society. But if so, this just indicates that the “communicative links” are yet to be established by which their social alterity would be relinquished in turn. The price of staying inside their own world is the maintenance of non-communication with the “outside world.” For the rest, wherever social modernity has unfolded its functional forms, no regional or state borders can impede their encompassing inclusion in world society, nor, as we will go on to discuss, their common neglect by function systems whose operations prove to be non-inclusive in actual practice.

FRACTIVERSE OR PLURIVERSE?

It sometimes seems that anthropologists are constitutionally averse to oneness, to singularity, and likewise obsessed with the plural. Never one world; always many worlds. Once these were the many worlds of symbolic culture; now, in the wake of the so-called “ontological turn,” we have the many worlds of essential being, of realities to be symbolized. Everyone and everything, it seems, is its own world. You name it, and there’s a world for you.

—Tim Ingold, “One World Anthropology”

Is the world posited by the conception of world society in social systems theory, which “provides one world for one system,” an extreme form of the “one-world world” that, according to sociologist and STS scholar John Law, has been the default ontology of modern Western thought? Law invokes this critique through an admittedly shorthand differentiation between “North” and “South,” roughly, between European metaphysics and Aboriginal cosmologies. Law notes that the English doctrine of *terra nullius* or “uninhabited land,” which licensed the expropriation of Aboriginal homelands in Australia, was underwritten

philosophically by a Northern mindset that regards the world as being “out there,” as self-subsistent, removed from human concerns, and thus indifferent to our behaviors regarding it. Regarding this notion of the world, for the North, “People do not perform it. It is outside us and we are contained inside it”; but for the South, “The idea of a reified reality out there, detached from the work and the rituals that constantly re-enact it, makes no sense.”⁹

According to Law, aside from the brutal power differential between colonial and native peoples, these divergent ideas about the world do not simply derive from a clash of archaic beliefs and modern convictions about how the world “really is.” It is not just that the North has accepted as the truth of nature a world apart from humans, which then licenses their domination of its resources, including its non- or allegedly less-than-human inhabitants:

If we say that Aboriginal people believe one thing and we believe something else then we may not realize that we are doing this, but in effect we are buying into a version of the *terra nullius* doctrine. This is because we are saying that the world (or even the *universe*) is really something like a large space-time box that goes on by itself [... and] we have not abandoned our basic commitment to the idea of a single all-encompassing reality.¹⁰

But there is more to it than that: for Law, the “one-world world” of Northern convictions covers over the reality of there being different realities produced by different practices, alternative enactments of different worlds. With a shift in ontological commitments, the singular Northern universe may be decomposed into a *fractiverse* of “*particular* worlds” that goes, so to speak, beyond belief:

If we live in a single container world, within a universe, in which nature has a definite and “natural” form, then we might imagine a liberal way of handling the power-saturated encounters between different kinds of people and our interpretations of the world. But if we live, instead, in a multiple world of different enactments, if we participate in a fractiverse, then there will be, there can be, no overarching logic or liberal institutions, diplomatic or otherwise, to mediate between the different realities. There is no “overarching.” Instead there are contingent, more or less local and practical engagements.¹¹

Now, broadly speaking, NST is oriented toward the European tradition as inflected by its adherents’ training or employment in the American academy and its cultivation within Western postwar developments in cybernetics and systems thinking. Luhmann in particular sees his work as a renovation or surpassing of “Old-European” thought styles, and his sociological writings do not typically invoke non-Western or aboriginal cultures. That said, the Western metaphysics under critique in the ANT wing of STS, on the one hand, and the deployment of observation and distinction in NST, on the other, do not overlap but generate radically different constructions of “the world.” At the risk of belaboring the obvious, we can note that Law’s mode of thought produces no explicit registration of functional differentiations, nor of the distinct regimes of communication these systemic differences generate. His discourse locks into the semantics of *reality*, *truth*, and *belief*, in terms of which it submits to the paradoxes built into these slippery tokens of logical unity. Thus, when confronted with flagrant heterogeneity in the distinctions different social systems use to produce divergent observations, in order to account for those differences, his discourse defaults to “different realities,” “different truths” in touch with some essential but particular being beyond the mere presentation of “different

beliefs.” What this discourse of the *fractiverse* does *not* observe is that the world is “an incomprehensible unity that can be observed in various ways, *and only in various ways*. Its ‘decomposition’ is not to be found, it can only be constructed, and this requires distinctions to be chosen.”¹² According to Luhmann’s scheme, then, whatever ontological claims may be made with regard to alternatives to the unity of the “world” or the “universe,” they rest on a choice of distinctions and produce what amount to oblique refractions of the open-ended multiplicity that always already inheres in the mundane differentiation of social systems.

So, no, the *world* of world society in social systems theory is not a version of the “one-world world” delineated by Law, precisely because it is in no way taken to be a self-subsistent or free-standing “reality.” Rather, it is an empty plenum necessarily generated as the outside of the autopoiesis of world society.

As Luhmann notes in *Theory of Society*,

In present-day thinking, the world is neither a beautiful living being nor an *aggregatio corporum*. Nor is it the *universitas rerum*, the totality of visible and invisible objects, things, and ideas. Nor, finally, is it infinity to be filled, absolute space or absolute time. It is not an entity that “contains” everything and therefore “lasts.” All these descriptions and many more are provided by the world. The world itself is only the overall horizon of all meaningful experience, whether directed inward or outward and forward in time or backward. It is not closed off by boundaries but by the meaning that can be activated within it. The world is to be understood not as an aggregate but a correlate of the operations taking place in it.¹³

By this light, an ontological construction like the *fractiverse* results from the repression of such epistemological reckoning in ANT and neighboring schools of thought such as speculative realism. Like everyone else, Law and his colleagues bring forth worlds, but they do not then acknowledge the self-referential nature of their critical operations. Rather, they displace the agency for observation, as in the article above, onto their indigenous interlocutors, or onto the Western scientific practitioners under the gaze of STS critique. The conceptual container that generates Law’s *fractiverse* is labeled “practice”: “Perhaps I need to add that there is nothing outside practice. We are embedded in practices all the way down.”¹⁴ But what is “practice” if not a mobile trope for the disciplining or operationalizing of semiotic or metrological technologies of observation in the service of fabrications?

What Law’s *fractiverse* of practices is gesturing toward may be more clearly indicated in anthropologist Arturo Escobar’s discourse of autonomy in relation to his articulation of the *pluriverse*, “*a world where many worlds fit*.”¹⁵ Escobar also couches his *pluriverse* as an ontological concept, but here, because it emerges through the discourse of autopoiesis in the systems theory of Maturana and Varela, the ontological framing is radically differential, relational, and so directed toward the constitution of interdependent autonomies, on the model of living systems. Escobar writes that

systems can undergo structural changes and adopt various structures in response to interactions with the environment, but they have to maintain a basic organization in order to remain as the units they are. Historical interaction among autopoietic units (worlds, one might say) often takes on a recurrent character, establishing a pattern of mutually congruent structural changes that allows the respective units to maintain their organization (pluriversal interactions). This eventually leads to the coordination of

behavior, communication, and social phenomena through co-ontogenies, resulting in all kinds of complex units (codesign); in humans, this process takes place through language.¹⁶

Perhaps it would not be too scandalous to suggest that *a world where many worlds fit*, an expression of political desire that Escobar cites from Mexico's Zapatista movement, could be rewritten in Luhmann's systemic idiom as "a world society encompassing many systems." Escobar's autopoietic discourse gains ontological weight by its immersion in issues of political autonomy for indigenous peoples, where life is literally precarious and put on the line in the sheer drive for collective survival and cultural continuation. Nevertheless, for better or worse, this particular community *has* "communicative links" (such as Escobar's writings) and so is tasked with making its way as a differentiated subsystem within world society.

AT THE MARGINS OF FUNCTION SYSTEMS

Escobar notes in his introduction that "ecologically oriented fields in particular have realized design's vital role in creating a more livable world."¹⁷ Escobar's own discourse could be described as a treatise on *world design*: "What the notion of design signals in this work—despite *design*'s multiple and variegated meanings—is diverse forms of life and, often, contrasting notions of sociability and the world."¹⁸ Escobar marshals the autopoietic systems theory of Maturana and Varela for social activism through a program of deliberate invention of and intervention in the structures available for worldly use. Luhmann's world society, in contrast, is a social systems-theoretical description of how modern society operates. Escobar's *design* would appear to fall under what Luhmann calls *planning*. But because social systems have their own mode of operational autonomy, their evolution is also self-produced: they cannot be *directly* steered by the conscious intentions, however admirable, of persons in their environment. What *is* possible, according to Luhmann, is a deliberate course of neocybernetic trial-and-error in the administration of designed incitements, in that "a self-referential system which tries to absorb planning may speed up its own evolution, because it becomes hypercomplex and will force itself to react to the ways in which it copes with its own complexity. If this is true, world society will have to face conditions in which more intentional planning will lead to more (and more rapid) unintentional evolution."¹⁹

Social systems theory does not aim to improve the world directly, but rather, indirectly. Its immediate goal is, as much as may be possible, "to improve the instruments of self-observation, i.e. of communicating within society about society"; applied to world society, "It is an international language, not designed to protect specific interest."²⁰ It seeks to see whether its "fascinating possibilities of arriving at a higher level of intelligibility" can be realized.²¹ What then is the picture of world society Luhmann draws in the last two decades of the twentieth century? Shortly before his death in 1998, Luhmann speculated that "inclusion and exclusion" would be "the guiding difference of the next century."²² Writ large, this looks very much like the world we currently inhabit, one that fulfills "the worst imaginable scenario" in operating by "the metacode of inclusion/exclusion."²³ As we readily observe in the environment of today's world society, "some human beings will be persons and others only individuals; ... some are included into function systems for (successful or unsuccessful) careers and others are excluded from these systems, remaining bodies that try to survive the next day."²⁴ By the early 1980s, Luhmann already noted that world society has "special risks" due to "self-defeating mechanisms For example, functional differentiation presupposes equality and

creates inequality.”²⁵ While function systems are nominally inclusive, since in the abstract, the economic subsystem or the political subsystem are open to all comers, nevertheless, in operation they place bouncers at the doors by establishing boundaries and processing differences (e.g., in the ability to make payments, in vote tallies, in truth claims, in test scores).

Luhmann’s writings labor assiduously to establish his description of world society as the most advanced formulation of its object. Yet into this scholarly industry a life experience fell that crystalized his own critique of the world system by opening a view onto the literal underside of its operations. In the account given by law professor and Luhmann scholar Andreas Philippopoulos-Mihalopoulos:

Luhmann discovered exclusion and with it his own *tristes tropiques* when in the early 1990s he visited the favelas in Brazil. The story is almost mythical in its recurrence, not least because of the spectacular new vocabulary that followed the journey. Expressions such as “existences reduced to the bodily ... attempting to get to the next day” in *Beyond Barbarism*, and “physical violence, sexuality, the elemental and impulsive satisfaction of necessities” and “the observation of bodies” in *Die Gesellschaft der Gesellschaft* have narrowly although indelibly opened the text to the observation of the hitherto unobservable.²⁶

By the later 1990s, then, Luhmann’s portrayal of how the autopoietic logic of function systems tilts them into exclusion and inequality had only gained in specificity:

Functional differentiation, as opposed to what the self-descriptions of the systems are claiming, cannot realize the postulated full inclusion. Function systems, when operating rationally, exclude persons or marginalize them so drastically that this has consequences in regard to getting access to other function systems. No education, no work, no income, no regular marriages, children with no birth certificate, no passport, no participation in politics, no access to legal advice, to the police, or to the courts—the list can be extended and it concerns, depending on the circumstances, all marginalizations up to total exclusion. No one will say that it has to be this way according to some kind of Malthusian population law. It is, however, sufficient to see that it is so and how such an amplifying effect is produced at the margins of function systems.²⁷

In this light, the current protests intensifying communication around structural racism in the United States would be the latest inflection of a more encompassing evolution of *systemic inequality*. Yes, “the system” is the problem, but where world society is concerned, there is no other system. We must hope somehow to condition its evolution. There is surely no quick fix. That there is *no* fix is unacceptable, but it surely will not be found by rounding up scapegoats for the guillotine. However, while social systems cannot be steered, they can be *disturbed* by individual psychic systems multiplying consciousness of grievance through resonant pulsations set up in the medium of meaning. “For only the consciousness of individuals is structurally coupled with the autopoiesis of the societal system,” writes Luhmann.²⁸ That is, events in the environment of meaning systems, in order to gain purchase within those systems, must be converted *into* meaning by those same systems, and “Only consciousness can irritate communication in a way that is compatible with the autopoiesis and the operational closure of the social system. All other environmental changes (physical, chemical, biological, e.g., death) can only have destructive effects,” that is, effects that threaten the continued autopoiesis of communication that is world society.²⁹

INCLUDING GAIA

I will conclude with an approach to the question whether “*all* other environmental changes” besides changes of mind “can only have destructive effects” on social systems, or whether it is possible to reverse the sign of this negation. Let us dwell a bit longer on this remarkable passage from the very late essay “Globalization or World Society.” Here we see some of the “spectacular new vocabulary” noted by Philippopoulos-Mihalopoulos, by which Luhmann’s own discourse registers a new level of ecological witness regarding “problems with fresh air and fresh water, with oil and with nourishment, with pollution and with the ozone layer depletion.”³⁰ Luhmann is unusually insistent in this passage on describing the processes of access that operate at the margins of conscious and social operations. After all, the systemic autonomy ensured by operational closure also produces cognitive access *to* the environment. These boundaries are indeed, so to speak, “semi-permeable”: they are open, not to direct transmissions, but to mutually resonant reconstructions within their own systemic orbits. The crucial contingency at hand is this:

Only consciousness can produce the noise necessary for the emergence and evolution of social order. Only conscious operations can perturbate the communicative system and create preconditions of sense-making within this system. Everything else—say, death, fire, earthquakes, climatic shifts, technological catastrophes—can only destroy communication. Such events can, of course, be observed, that is, thematized by the social system, but to do so requires communication and, as its external condition, consciousness.³¹

Let us place these remarks in the context of the planetary environment of world society. Thus, the autopoiesis of consciousness by individuals must first process the *ecological* noise to which social systems are constitutionally deaf. This is the necessary preparation enabling its reconstruction by the autopoiesis of communication in social systems such that ecological *meaning* and the train of its implications percolate out to world society. In this circuit, Luhmann declares, “the distinction between human and ecological conditions is quite an artificial construction, reflecting the internal operative necessities of society, whereas individuals as far as their organic life is concerned, are, in fact, part of the ecological environment, contributing to and suffering from its deterioration.”³² This reconnection of mind to body would suggest that world society *can* be brought to cognize climate change as a threat to the continuation of its *own* ecology, constituted by conscious individuals *and* the wider living world *and* the Earth system as well. And although it does not register explicitly in Luhmann’s text, the idea of Gaia provides an efficient shorthand notation for the messages demanding communication within the “new climatic regime.” Let us say that Gaia has now become *too noisy to be neglected any longer* by the function systems, too insistent to be excluded from assuming an address within world society. We may take philosopher of science Isabelle Stengers’s influential articulation of “Gaia the Intruder” as a virtual model for this operation of ecological sense-making:

We are no longer dealing (only) with a wild and threatening nature, nor with a fragile nature to be protected, nor a nature to be mercilessly exploited. The case is new. Gaia, she who intrudes, *asks nothing of us*, not even a response to the question she imposes. Offended, Gaia is indifferent to the question “who is responsible?” and doesn’t act as a righter of wrongs—it seems clear that the regions of the earth that will be affected first will be the poorest on the planet, to say nothing of all those living beings that have nothing to do with the affair.³³

Of course, it is we humans, not Gaia, who operate in the medium of meaning through linguistic codes that make it possible to pose questions about human responsibility for environmental deterioration. Perhaps the Gaian system is as indifferent to what it excludes from *its* operations as is any other autopoietic system. At the same time, like an inverted reflection of world society, Gaia too cannot be constrained by the borders of nation states or by the identity formations of ethnicities, cultures, or other “species-theoretical terms.”³⁴ Here again, speculating upon societal evolution from his particular systemic vantage, Luhmann saw coming clearly enough the place where we are now: “That the ecological impacts of societal operations cannot be restricted to regional territories does not need any further argument. But their impact on individuals, too, seems to be a universal phenomenon and this will become more evident in the near future.”³⁵ But are we humans, as far as our organic life is concerned, and as the agents of the thing called the Anthropocene epoch, not in fact *included* in Gaia’s operations as well? Shall we not take Gaia’s intrusion as an inarticulate indication of the onset of potentially catastrophic climate change? If so, perhaps Gaia theory’s concerted relay of this message could provoke a potentially constructive mutation in our function systems. Stranger things have happened.

To complete our final turn toward Gaia as a counterpart to world society, we can consider how the work of the evolutionary theorist Lynn Margulis renders the formal but empty category of *world* in social systems theory distinct through a discourse of Gaia as the planetary autopoietic system. I cannot open up this discussion here but must refer the interested reader to my treatments of Margulis in neocybernetic context in *Gaian Systems*. In short, the comparison is now between two different *systems-theoretical* worlds. One is the world we have explored in this chapter, understood epistemologically and in its simultaneity as the indistinct or unmarked environment beyond the autopoietically closed operations of psychic and social systems. The other is an autopoietic conception of the Gaian system itself, emerging in cosmological time from planetary couplings of physical, material-energetic dynamics to self-producing biological and ecological processes. In his late writings at least, some of which we have reviewed here, Luhmann appears to have realized that by its move away from embodiment toward the virtual dynamics of consciousness and communication, by excluding the living world, consigning it solely to the indifferent environment of meaning systems, social systems theory paid for its gain in analytical lucidity with a loss in material, worldly purchase. As developed by Margulis in particular, Gaia theory vigorously *worlds* neocybernetic systems theory once again. By looping metabiotic autopoietic conceptuality toward a planetary consortium that renders the distinction between system and environment even more complex, NST now restores the “unity of their distinction” on a level that itself encompasses world society.

NOTES

- 1 See my entries in References.
- 2 Niklas Luhmann, “The World Society as a Social System,” *International Journal of General Systems* 8 (1982): 131.
- 3 Luhmann, *Social Systems*, trans. John Bednarz Jr. with Dirk Baecker (Stanford, CA: Stanford University Press, 1995), 4.
- 4 Luhmann, “World Society,” 131.
- 5 See Elena Esposito, “Artificial Communication? The Production of Contingency by Algorithms,” *Zeitschrift für Soziologie* 46.4 (2017): 249–265.

- 6 Luhmann, "World Society," 132.
- 7 Ibid., 132–133.
- 8 Luhmann, *Theory of Society*, Volume 1, 83. For more discussion of Luhmann's world society, see the exposition in Hans-Georg Moeller, *Luhmann Explained: From Souls to Systems* (Chicago: Open Court, 2006), 52–63, and the critical analysis in Andreas Philippopoulos-Mihalopoulos, *Niklas Luhmann: Law, Justice, Society* (New York: Routledge, 2010), 116–28.
- 9 John Law, "What's Wrong with a One-World World?" *Distinktion: Scandinavian Journal of Social Theory* 16.1 (2015): 126–127.
- 10 Ibid., 127.
- 11 Ibid., 127–128, 136.
- 12 Luhmann, *Theory of Society*, Vol. 1, 90; my italics.
- 13 Ibid., 88–89.
- 14 Law, "One-World World," 130.
- 15 Arturo Escobar, *Designs for the Pluriverse: Radical Interdependence, Autonomy, and the Making of Worlds* (Durham: Duke University Press, 2018), xvi.
- 16 Ibid., 170.
- 17 Ibid., 2.
- 18 Ibid., 3.
- 19 Luhmann, "World Society," 134.
- 20 Ibid., 137.
- 21 Ibid.
- 22 Luhmann, "Beyond Barbarism," trans. Hans-Georg Moeller, in *Luhmann Explained: From Souls to Systems* (Chicago: Open Court, 2006), 270.
- 23 Luhmann, "Globalization or World Society: How to Conceive of Modern Society?" *International Review of Sociology* 7.1 (1997): 76
- 24 Ibid., 76.
- 25 Luhmann, "World Society," 134.
- 26 Philippopoulos-Mihalopoulos, *Niklas Luhmann*, 122–123.
- 27 Luhmann, "Beyond Barbarism," 270.
- 28 Luhmann, "Globalization or World Society," 72.
- 29 Ibid., 72.
- 30 Ibid., 73.
- 31 Ibid.
- 32 Ibid.
- 33 Isabelle Stengers, *In Catastrophic Times: Resisting the Coming Barbarism*, trans. Andrew Goffey (London: Open Humanities Press, 2015), 46.
- 34 Luhmann, *Theory of Society*, Vol. 1, 94.
- 35 Luhmann, "Globalization or World Society," 74.

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